

Literature Status Note: Arveson’s Problem D

Run `fa_banach_001` — lane 5

9 August 2026

Status: `literature_already_answered`.

Original question

William Arveson’s *p-Summable Commutators in Dimension d* (arXiv:math/0308104; *J. Operator Theory* 54 (2005), 101–117) asks in Problem D on PDF page 16, in the concluding section:

Let $\bar{T}$ be a finite-rank d -contraction. Is $\bar{T}$ a Fredholm d -tuple? Does the index formula hold?

The immediately preceding sentence makes clear that the intended class is finite-rank *pure* d -contractions. The index formula is the identity relating Arveson’s curvature invariant $\kappa(T)$ to the Fredholm index, $\kappa(T) = (-1)^d \operatorname{ind}(T)$.

Identification of the answer

Gleason, Richter, and Sundberg prove in Theorem 1.3 (restated as Theorem 4.5) that for every pure finite-rank d -contraction,

$$\kappa(T) = (-1)^d \operatorname{ind}(T - \lambda) \quad (\lambda \in \mathbb{B}_d \setminus \sigma_e(T)).$$

In particular, if T itself is Fredholm, then the formula requested by Problem D holds at $\lambda = 0$.

The example following Theorem 4.3 (supporting PDF pages 21–22) constructs an invariant subspace $M \subset H_2^2$ for which $(M_{z_1} - \lambda, M_{z_2})|_M$ is not Fredholm at any $(\lambda, 0)$ in its zero set. Via the standard compression model, Theorem 4.5 then yields pure finite-rank 2-contractions with $0 \in \sigma_e(T)$, hence non-Fredholm examples. The introduction states this consequence explicitly on supporting PDF page 6 and points directly to “Problem D of [12]”, with [12] being Arveson’s paper.

Therefore the universal Fredholm assertion in Problem D is false for $d > 1$, whereas its index formula holds for every tuple in the stated class that is Fredholm. This is a complete literature answer to Problem D.

Scope limitation

This identification does not settle Arveson’s Conjectures A–C concerning polynomially generated invariant subspaces, compact commutators, essential normality, or their more specialized quotient modules.

Evidence and search bound

The source question was checked against the local arXiv source and PDF. The supporting paper was found by exact-title, “Problem D”, “finite rank pure d -contraction”, and “not Fredholm” searches. Its introduction, Theorems 1.3 and 4.5, the example following Theorem 4.3, and bibliography entry [12] were checked directly. The explicit citation and answer make this a provenance record, not a new mathematical result.

References

- [1] W. Arveson, *p*-Summable commutators in dimension d , J. Operator Theory 54 (2005), 101–117; arXiv:math/0308104.
- [2] J. Gleason, S. Richter, and C. Sundberg, *On the index of invariant subspaces in spaces of analytic functions of several complex variables*, J. Reine Angew. Math. 587 (2005), 49–76; doi:10.1515/crll.2005.2005.587.49.